Nathan Nicholas D'Souza

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SKILLS & INTERESTS

- **AI Engineering & Machine Learning:** LangGraph, Hybrid RAG (BM25 + Dense RRF), Semantic Caching, Quantized ONNX (FastEmbed), PyTorch, TensorFlow, Keras, OpenCV, Pytesseract (OCR), LLM API Orchestration, Guardrails & Eval Harnesses, Computer Vision.
- **MLOps, Data & Infrastructure:** Docker, Docker Compose, PostgreSQL (pgvector), Redis, SQL (SQLite), CI/CD (GitHub Actions), PyTest, Vitest, Playwright, ETL Pipelines, Power BI, NumPy, Pandas, RStudio, Openpyxl.
- **Full-Stack Development:** Python (FastAPI, Django, Flask, PySide6), TypeScript & JavaScript (Next.js, React, React Native), Server-Sent Events (SSE), Pydantic, Tailwind CSS, Java, C/C++, RESTful APIs, Agile/Scrum (Jira), Figma.

EXPERIENCE

Assistant, HR Consulting Services | *Fraser Health Authority - Surrey, Canada* *Sept 2025 – Present*

- **Architected full-stack SDLC solutions** using React Native/Django/SQLite to implement digital phase-tracking and automated paper trails for 100% of agency contracts.
- **Developed ETL-driven dashboards** with Django and Chart.js to visualize 8-week staffing trends, eliminating high-latency manual data retrieval from Outlook.
- **Optimized legacy automation suites** in PySide6, updating script logic to ensure 100% reliability for critical departmental workflows and real-time data accessibility.

Workforce QI and Optimization Co-op | *Fraser Health Authority - Surrey, Canada* *Aug 2024 – Aug 2025*

- **Developed ETL pipelines** via a PySide6/pywin32 RFP Utility to automate invoice extraction and organization, reducing weekly manual labor from **30+ hours to under 5 minutes.**
- **Accelerated UKG WFM migration by 60%** through Python-based audit automation and custom PDF-to-Excel statement reporting for complex vendor payment tracking.
- **Engineered Power BI analytics** for workforce scheduling and payroll, performing regression-style checks to ensure data integrity and equitable employee shift distribution.

PROJECTS

Aegis AI Race | Production AI Platform vs. Naive Wrapper *Sept 2026 - Oct 2026*

- **Architected a full-stack AI benchmarking platform** (Next.js, FastAPI, LangGraph) streaming real-time Server-Sent Events (SSE) telemetry to race a multi-stage RAG pipeline against a baseline 70B LLM.
- **Engineered Hybrid RAG** (PostgreSQL pgvector + BM25 via Reciprocal Rank Fusion) and Redis Semantic Caching, achieving 7.3 ms cache hits, 28 ms retrieval, and 100% elimination of invoice audit hallucinations.
- **Containerized via Docker/Compose** and validated reliability using **150 automated tests** (PyTest, Vitest, Playwright E2E) across a GitHub Actions CI pipeline.

Clarity | AI Time Awareness Assistant | *Link* *Sept 2025 – Nov 2025*

- **Engineered predictive modeling** via an AI feedback loop (Claude API) that learns from historical performance data to generate accurate task estimates.
- **Led cross-functional Agile workflows** as the sole developer using Jira/Discord to build a React Native/Django MVP with Google Calendar API integration.

BinIt! | An AI-Powered Trash Sorter | *Link* *March 2023*

- Engineered BinIt! backend with **Flask** and frontend with **ReactJS**, employing **base64** encoding for image transfer, while leveraging the **ResNet152** model for object detection, enhancing sorting accuracy

EDUCATION

Simon Fraser University | *BSc in Computing Science, Conc. Artificial Intelligence* *Sept 2020 – Dec 2025*

- **Relevant Coursework**: Machine Learning, Artificial Intelligence, Database Systems, Computer Vision, Data Structures & Algorithms, Software Engineering.

AWARDS & ACHIEVEMENTS

Mountain Madness 2025: Placed **1st** as *"The Classic"* Hackathon Winners, earning CAD **200** *March 2025*
Opportunity Fest 2023: Awarded **CAD 250** for securing **2nd** place in **Social Innovation** as HelpingHands *November 2023*
SystemHacks 2023: *Achieved the **Best Project** in the category WiCS Award for our project **'BinIt!'*** *March 2023*